

```
import pandas as pd
import numpy as np

from sklearn.feature_extraction.text import CountVectorizer
from sklearn.metrics.pairwise import cosine_similarity
```

```
# Load dataset
df = pd.read_csv("Dataset .csv")

# Display first rows
df.head()
```

	Restaurant ID	Restaurant Name	Country Code	City	Address	Locality	Locality Verbose	Longitude	Latitude	Cuisines	...	Currency	Has Table booking	Has Online delivery	Is delivering now	Switch to order menu	Price range	Aggregate rating	Rating color	Rating text	Votes	
0	6317637	Le Petit Souffle	162	Makati City	Third Floor, Century City Mall, Kalayaan Avenu...	Century City Mall, Poblacion, Makati City	Century City Mall, Poblacion, Makati City, Mak...	121.027535	14.565443	French, Japanese, Desserts	...	Botswana Pula(P)	Yes	No	No	No	3	4.8	Dark Green	Excellent	314	
1	6304287	Izakaya Kikufuji	162	Makati City	Little Tokyo, 2277 Chino Roces Avenue, Legaspi...	Little Tokyo, Legaspi Village, Makati City	Little Tokyo, Legaspi Village, Makati City, Ma...	121.014101	14.553708	Japanese	...	Botswana Pula(P)	Yes	No	No	No	3	4.5	Dark Green	Excellent	591	
2	6300002	Heat - Edsa Shangri-La	162	Mandaluyong City	Edsa Shangri-La, 1 Garden Way, Ortigas, Mandal...	Edsa Shangri-La, Ortigas, Mandaluyong City	Edsa Shangri-La, Ortigas, Mandaluyong City, Ma...	121.056831	14.581404	Seafood, Asian, Filipino, Indian	...	Botswana Pula(P)	Yes	No	No	No	4	4.4	Green	Very Good	270	
3	6318506	Ooma	162	Mandaluyong City	Third Floor, Mega Fashion Hall, SM Megamall, O...	SM Megamall, Ortigas, Mandaluyong City	SM Megamall, Ortigas, Mandaluyong City, Mandal...	121.056475	14.585318	Japanese, Sushi	...	Botswana Pula(P)	No	No	No	No	4	4.9	Dark Green	Excellent	365	
4	6314302	Sambo Kojin	162	Mandaluyong City	Third Floor, Mega Atrium, SM Megamall, Ortigas...	SM Megamall, Ortigas, Mandaluyong City	SM Megamall, Ortigas, Mandaluyong City, Mandal...	121.057508	14.584450	Japanese, Korean	...	Botswana Pula(P)	Yes	No	No	No	4	4.8	Dark Green	Excellent	229	

5 rows × 21 columns

```
# Select useful columns
```

```
data = df[['Restaurant Name',
'Cuisines',
'City',
'Price range',
'Aggregate rating'
]]

data.head()
```

	Restaurant Name	Cuisines	City	Price range	Aggregate rating	
0	Le Petit Souffle	French, Japanese, Desserts	Makati City	3	4.8	
1	Izakaya Kikufuji	Japanese	Makati City	3	4.5	
2	Heat - Edsa Shangri-La	Seafood, Asian, Filipino, Indian	Mandaluyong City	4	4.4	
3	Ooma	Japanese, Sushi	Mandaluyong City	4	4.9	
4	Sambo Kojin	Japanese, Korean	Mandaluyong City	4	4.8	

Next steps:

[Generate code with data](#)

[New interactive sheet](#)

```
# Fill missing values
```

```
data['Cuisines'] = data['Cuisines'].fillna('Unknown')
data['City'] = data['City'].fillna('Unknown')
```

```
/tmp/ipykernel_3584/2864030711.py:3: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
data['Cuisines'] = data['Cuisines'].fillna('Unknown')
/tmp/ipykernel_3584/2864030711.py:4: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
data['City'] = data['City'].fillna('Unknown')
```

```
# Combine features into single column
```

```
data['Features'] = (
    data['Cuisines'] + " " +
    data['City'] + " " +
    data['Price range'].astype(str)
)

data.head()
```

```
/tmp/ipykernel_3584/3974754825.py:3: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
data['Features'] = (

```

	Restaurant Name	Cuisines	City	Price range	Aggregate rating	Features	
0	Le Petit Souffle	French, Japanese, Desserts	Makati City	3	4.8	French, Japanese, Desserts Makati City 3	
1	Izakaya Kikufuji	Japanese	Makati City	3	4.5	Japanese Makati City 3	
2	Heat - Edsa Shangri-La	Seafood, Asian, Filipino, Indian	Mandaluyong City	4	4.4	Seafood, Asian, Filipino, Indian Mandaluyong C...	
3	Ooma	Japanese, Sushi	Mandaluyong City	4	4.9	Japanese, Sushi Mandaluyong City 4	
4	Sambo Kojin	Japanese, Korean	Mandaluyong City	4	4.8	Japanese, Korean Mandaluyong City 4	

Next steps:

[Generate code with data](#)

[New interactive sheet](#)

```
# Convert text data into matrix
```

```
cv = CountVectorizer()

feature_matrix = cv.fit_transform(data['Features'])
```

```
# Compute similarity scores
```

```
similarity = cosine_similarity(feature_matrix)

print(similarity)
```

```
[1. 0.77459667 0.18257419 ... 0. 0. 0. ]
[0.77459667 1. 0.23570226 ... 0. 0. 0. ]
[0.18257419 0.23570226 1. ... 0. 0. 0. ]
...
[0. 0. 0. ... 1. 0.28867513 0.35355339]
[0. 0. 0. ... 0.28867513 1. 0.81649658]
[0. 0. 0. ... 0.35355339 0.81649658 1. ]]
```



```
def recommend_restaurant(name):  
  
    index = data[data['Restaurant Name'] == name].index[0]  
  
    scores = list(enumerate(similarity[index]))  
  
    sorted_scores = sorted(  
        scores,  
        key=lambda x: x[1],  
        reverse=True  
    )  
  
    recommended = sorted_scores[1:11]  
  
    recommendations = []  
  
    for item in recommended:  
  
        restaurant_index = item[0]  
  
        restaurant = data.iloc[restaurant_index]  
  
        recommendations.append([  
            restaurant['Restaurant Name'],  
            restaurant['Cuisines'],  
            restaurant['City'],  
            restaurant['Aggregate rating']  
        ])  
  
    recommendation_df = pd.DataFrame(  
        recommendations,  
        columns=[  
            'Restaurant',  
            'Cuisine',  
            'City',  
            'Rating'  
        ]  
    )  
  
    return recommendation_df
```

data['Restaurant Name'].head(20)

	Restaurant Name
0	Le Petit Souffle
1	Izakaya Kikufuji
2	Heat - Edsa Shangri-La
3	Ooma
4	Sambo Kojin
5	Din Tai Fung
6	Buffet 101
7	Vikings
8	Spiral - Sofitel Philippine Plaza Manila
9	Locavore
10	Silantro Fil-Mex
11	Mad Mark's Creamery & Good Eats
12	Silantro Fil-Mex
13	Guevarra's
14	Sodam Korean Restaurant
15	Cafe Arabelle
16	Nonna's Pasta & Pizzeria
17	Balay Dako
18	Hobing Korean Dessert Cafe
19	Wildflour Cafe + Bakery

dtype: object

```
recommend_restaurant("Cafe Coffee Day")
```

	Restaurant	Cuisine	City	Rating	
0	ToozY	Cafe	Faridabad	0.0	
1	Cafe Bite	Cafe	Faridabad	3.2	
2	Cafe Coffee Day	Cafe	Faridabad	3.3	
3	The Binge Box Cafe	Cafe	Faridabad	3.4	
4	Tmos Cafe Corner	Cafe	Faridabad	0.0	
5	The Chaiwalas	Cafe	Faridabad	0.0	
6	Tamasha In Tafree	Cafe	Faridabad	0.0	
7	The Chocolate Room	Cafe, Desserts	Faridabad	3.6	
8	The Chai Cafe	Cafe, Chinese	Faridabad	3.1	
9	Funk House Cafe	Cafe, Italian, Salad	Faridabad	2.6	

Insights

1. Restaurants with similar cuisines and price ranges were grouped together.
2. Content-based filtering successfully generated personalized recommendations.
3. Cuisine type played a major role in recommendation similarity.
4. Combining multiple restaurant features improved recommendation quality.

Conclusion

This project developed a content-based restaurant recommendation system using restaurant features such as cuisines, city, and price range.

The recommendation engine used:

- Text vectorization
- Cosine similarity
- Feature combination

The system successfully recommended restaurants similar to user preferences.